

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Term: Quiz 1 (SET 1)

Course Number: Math 4241

Course Title: Integral Calculus and Differential Equations

Full Marks: 15

Time: 30 Minutes

Answer all the questions. The symbols have their usual meanings. Marks of each question are written in the brackets.

1. a)	Define inscribed rectangle. Find the upper and lower sums for the region bounded by the graph of $f(x) = x^2$ and the x-axis between $x = 0$ and $x = 2$.	(1+3+3)
b)	Define Reimann sum. Derive the reduction formula for $\sin^n x$.	(2+3)
	Define Gamma and Beta functions.	(2)
c)	Write down the order and degree of the following differential equations, also mention if the differential equation is linear or non-linear. (i) $(x+1) \frac{dy}{dx} - y^2 = e^x(x+1)^2$ (ii) $(x^4 + 2) \frac{d^2y}{dx^2} + (3x^2 - x) \frac{dy}{dx} + y = 0$ (iii) $\left(\frac{d^3y}{dx^3}\right)^2 + x \frac{d^2y}{dx^2} = x^2$ (iv) $\frac{d^3y}{dx^3} + x \frac{d^2y}{dx^2} + x \frac{dy}{dx} + 2y = e^x$	(1+1+1+1)
d)	Define non-linear DE.	(2)

Name:**Student ID:**

(Start writing from the line below)